7(a)	Since F = - dEp/dx & dEp/dx is the gradient of graph. Hence force is proportional to the gradient of the curve.	[1] [1]
(b)(i)	Since all values of E_p are negative (Fig 7.2), & $E_P = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{r}$ E_p is negative only if the 2 pt charges are of opposite sign. or F = -dE_p/dx Since dE_p/dx is positive, (fr Fig 7.2), F is negative. Since a negative F indicates an attractive force between the charges, the charges are of opposite sign.	[1] [1]
(ii)	When Q_p is doubled: Since $E_P = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{r}$, E_p is doubled at every x. Thus gradient would be doubled/increased/become steeper. or, since F = Q_pQ_Q / (4□$\sum_0 x^2$), F is doubled for every x. & F = - gradient of graph, gradient is doubled at every x.	 [1] [1]
(c)	$E_p = \frac{(charge\ of\ P)(charge\ of\ Q)}{4\pi\varepsilon_o r}$ Since (charge of P) x (charge of Q) = constant E_p 1/r E_p = - 3.6 eV at x = 4 x 10^{-10} m E_p = - 5.1 eV at x = r 3.6/5.1 = r / 4 x 10^{-10} r = 2.82 x 10^{-10} m	 [1] [1]

8 (a)(i)		